

<Practical Questions for ECMT 675 TA session>¹⁾

- 1.) (20 points) Consider the following equations:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

$$\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

Here $\mathbf{y}$ is a $(T \times 1)$ vector of observations on the dependent variable, $\mathbf{X}$ is a $(T \times k)$ matrix of observations on the explanatory variables with rank k , and $\boldsymbol{\beta}$ is the true value of a $(k \times 1)$ vector of parameters. Let b_2 denote the second element of the vector $\mathbf{b}$ and let β_2 denote the second element of the vector $\boldsymbol{\beta}$. Consider the following theorem:

Let $\hat{\beta}_2$ be any unbiased estimator of β_2 that is constructed from some function of $\mathbf{X}$ and $\mathbf{y}$. If condition A holds, then $\text{Var}(\hat{\beta}_2|\mathbf{X}) \geq \text{Var}(b_2|\mathbf{X})$.

- a.) Write down an equation for what it means to say that $\hat{\beta}_2$ is an unbiased estimator of β_2 .
b.) State a sufficient condition or conditions A for which the above theorem is true.

- 2.) (30 points) Suppose that $y_t \sim \text{i.i.d. } N(\mu, \sigma^2)$ with $\mu \neq 0$ and let $q_T = T / \sum_{t=1}^T y_t$. Prove that $\sqrt{T}(q_T - \mu^{-1}) \xrightarrow{L} N(0, v)$ and calculate v . Does $v = \lim_{T \rightarrow \infty} E(q_T - \mu^{-1})^2$?

1) This problem set comes from Dr. James Hamilton(UCSD) mid-term.

3.) (90 points total) This question asks you to examine the consequences of measurement error in the explanatory variable. Suppose that there is a relation between y_t and a scalar z_t of the form

$$y_t = \beta z_t + u_t.$$

The economist unfortunately does not have data on z_t , but instead has observation on x_t , which differs from the true z_t by a measurement error term v_t :

$$x_t = z_t + v_t.$$

Let $\mathbf{q}_t = (z_t, u_t, v_t)'$ and suppose that $\mathbf{q}_t$ is i.i.d. multivariate Normal:

$$\mathbf{q}_t \sim N \left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_z^2 & 0 & 0 \\ 0 & \sigma_u^2 & 0 \\ 0 & 0 & \sigma_v^2 \end{bmatrix} \right). \quad (1)$$

Note that this implies that the relation between y_t and x_t is given by

$$y_t = \beta x_t + \varepsilon_t \quad (2)$$

where

$$\varepsilon_t = u_t - \beta v_t \quad (3)$$

$$x_t = z_t + v_t. \quad (4)$$

a.) (10 points) Does $y_t = x_t\beta + \varepsilon_t$ satisfy the Gaussian regression model? Explain.

b.) (40 points) Let b denote the OLS estimator

$$b = \left(\sum_{t=1}^T x_t^2 \right)^{-1} \left(\sum_{t=1}^T x_t y_t \right).$$

Use equations (1)-(4) to calculate the plim of b .

c.) (10 points) Let $\beta^* = \text{plim } b$. Under what circumstances would β^* rather than β be the object that a researcher wanted to estimate?

d.) (20 points) Find the asymptotic distribution of $\sqrt{T}(b - \beta^*)$. Hint: if $a_t \sim N(0, \sigma^2)$, then $E(a_t^4) = 3\sigma^4$.

e.) (10 points) Simplify your answer to (d) for the special case when there is no measurement error ($\sigma_v^2 = 0$). Hint: you should be able to get the right answer for this even if you were completely lost for (a)-(d).